NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

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Department : Computer Science and Engineering

TOPIC: Software Quality management



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Introduction

- How is a quality product traditionally defined?
- Fitness of purpose:
  - **A quality product does exactly what the users want it to do.**

Fitness of Purpose

- What is fitness of purpose for software products?
  - **Satisfaction of the requirements specified in SRS document…**

"Quality is conformance to requirements"

-- Philip Crosby, "Quality is Free" 1979

Fitness of Purpose

- Basically user's view…
- A satisfactory definition of quality for many products:
  - A car, a table fan, a food mixer, microwave oven, etc.
- But, not satisfactory for software products.
  - **Why?**

Quality for Software Products

- Consider a software product:
  - Functionally correct:
    - Performs all functions as specified in the SRS document.
  - But, has an almost unusable user interface.
    - **Would you consider it as a quality product?**

Quality for Software Products

- Consider another example:
  - A product which does everything that users want.
  - But has an incomprehensible and unmaintainable code.
  - **Will you call it a quality product?**
- **Quality of a product needs to described in terms of a set of quality attributes.**

Product Quality Attributes

- Attributes of a good software:
  - **Correctness**
  - **Reliability**
  - **Efficiency:** Software should not make wasteful use of system resources (disk and memory space, CPU time, etc.) and should satisfy response time requirements
  - **Portability**
  - **Usability**
  - **Reusability**
  - **Maintainability**

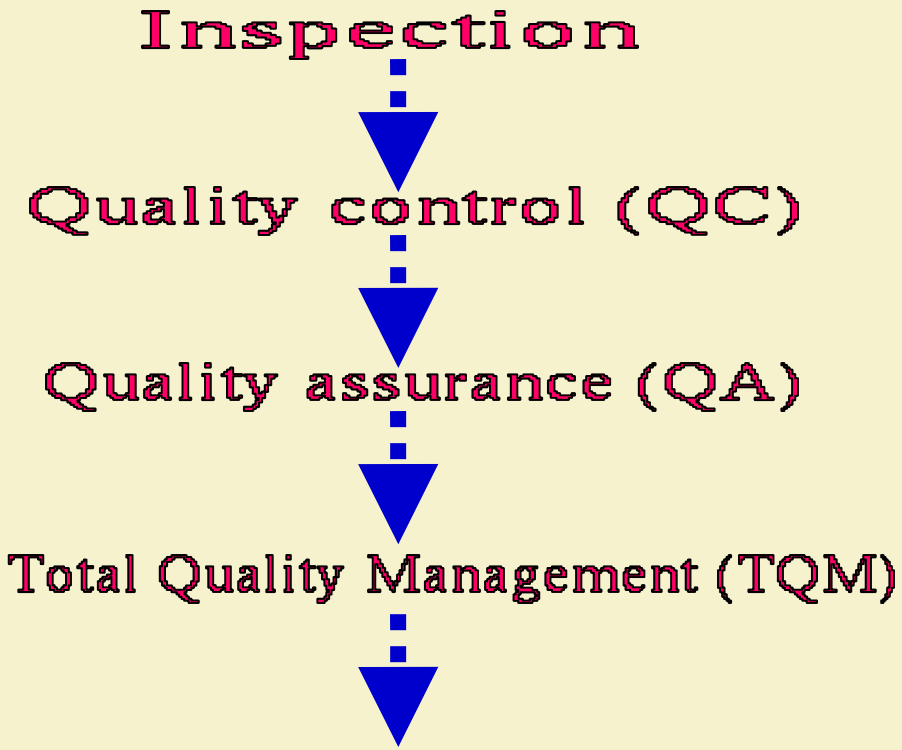
Evolution of Quality Systems

- Quality systems have evolved:
  - Over the last six decades.
- Prior to World War II:
  - Accepted way to produce quality products:
    - **Inspect the finished products**
    - **Eliminate defective products.**

Evolution of Quality System

- Since World war II,
  - Quality systems of organizations have undergone:
    - Four stages of evolution.
- Many advances came from Japanese:
  - Helped resurrect Japanese economy.

Evolution of Quality System



Product assurance

Process assurance

Evolution of Quality Systems

- Initial product inspection method:
  - Gave way to quality control (QC).
- **Quality control – Basic principle:**
  - **Not only detect the defective products and eliminate them**
  - **But also determine the causes behind the defects.**

Quality Control (QC)

- Quality control aims at correcting the causes of errors:
  - Not just rejecting defective products.
- **Statistical quality control (SQC):**
  - Inspection or testing of all products is cumbersome and many times not feasible.
  - **Quality of the output of the process is inferred using statistical methods.**

Quality Control (QC)

- The next breakthrough:
  - Development of **quality assurance principles.**

Quality Assurance

- Basic premise of modern quality assurance:
  - **If an organization's processes are good and are followed rigorously:**
    - **The products are bound to be of good quality.**

Quality Assurance

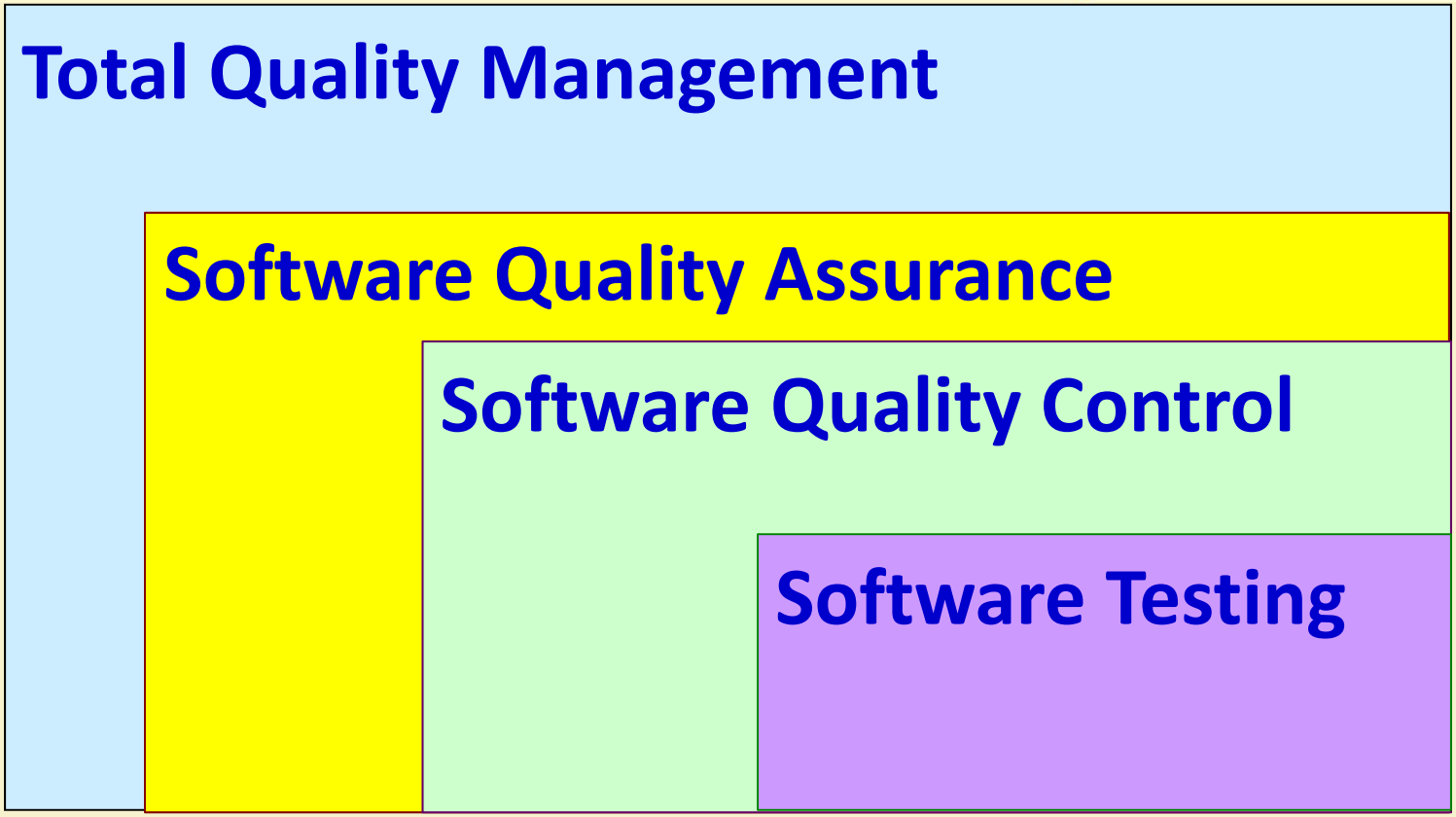
- All modern quality paradigms include:
  - Guidance for recognizing, defining, analyzing, and improving the production process.

Total Quality Management (TQM)

- TQM advocates:
  - **Continuous process improvements through process measurements.**

Total Quality Management (TQM)

- TQM goes beyond documenting processes
  - Optimizes them through redesign.
- Over the years the quality paradigm has shifted:
  - **From product assurance to process assurance.**


Total Quality Management
Software Quality Assurance
Software Quality Control
Software Testing

Process Improvement

- Implies introducing process changes to improve:
  - **Product quality**
  - **Reduce costs**
  - **Accelerate schedules.**

Process and Product Quality

- A good process is usually required to produce a good product.
- **For manufactured goods, process is the principal quality determinant.**
- For design-based activity, other factors are also involved:
  - For example, the capabilities of the designers.

ISO 9000

- ISO (international Standards Organization):
  - A consortium of 63 countries established to formulate and foster standardization.
- ISO published its 9000 series of standards in 1987.

ISO 9000

- A set of guidelines for the production process.
  - **Not directly concerned about the product itself.**
  - A series of three standards:
    - **ISO 9001, ISO 9002, and ISO 9003.**

ISO 9000

- Based on the premise:
  - **If a good production process is followed:**
    - **Good quality products are bound to follow.**

ISO 9001

- Applies to:
  - Organizations engaged in design, development, production, and servicing of goods.
  - **Applicable to most software development organizations.**

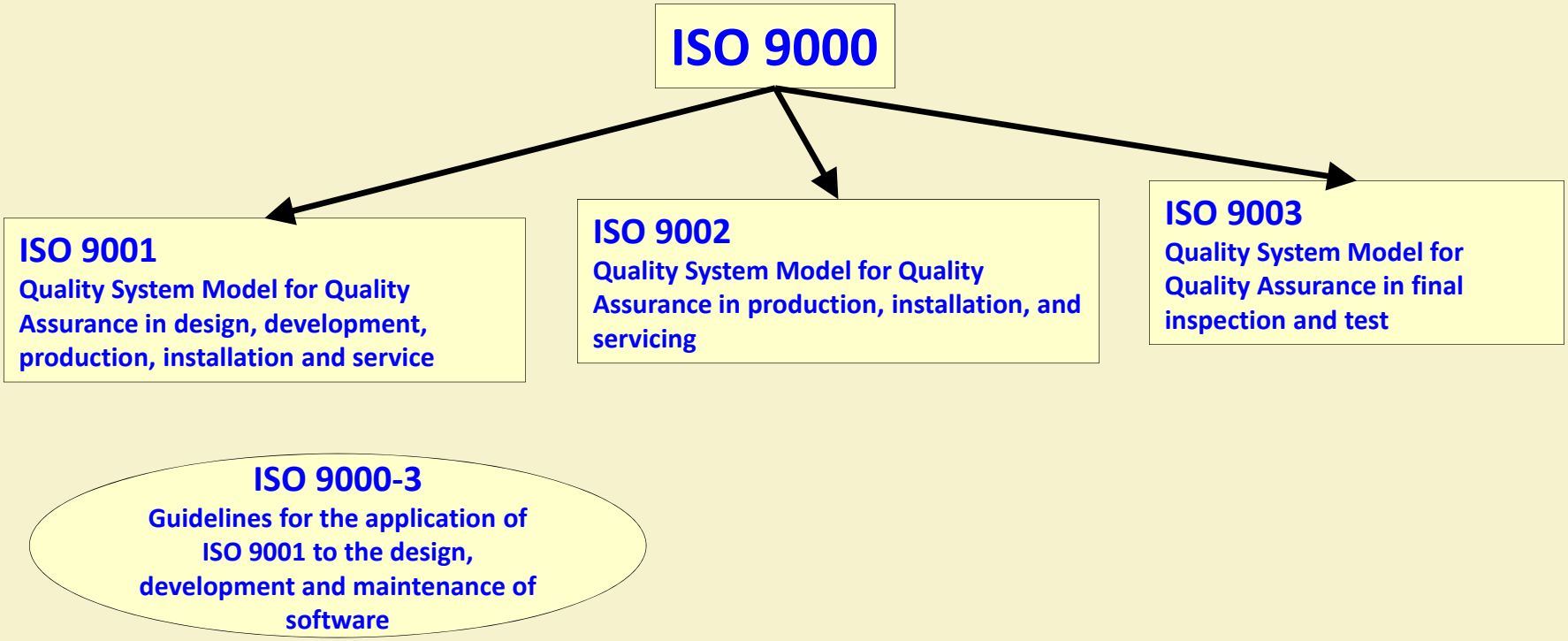
ISO 9002

- ISO 9002 applies to:
  - **Organizations who do not design products:**
    - **but are only involved in production, servicing.**
- Examples of this category of industries:
  - Steel or car manufacturing industries
  - Buy the product and plant designs from external sources:
    - only manufacture products.
  - **Not applicable to software development organizations.**

ISO 9003

- ISO 9003 applies to:
  - Organizations involved only in inspection and testing of products.

ISO 9000 Structure



ISO 9000 for Software Industry

- **ISO 9000 is a generic standard:**
  - Applicable to many industries,
    - Starting from a steel manufacturing industry to a service rendering company.
- Many clauses of ISO 9000 documents:
  - Use generic terminologies
  - **Very difficult to interpret them in the context of software organizations.**

Software vs. Other Industries

- Very difficult to interpret many clauses for software industry:
  - Software development is radically different from development of other products.

Software vs. Other Industries

- During software development:
  - **The only raw material consumed is data.**
- For any other product development:
  - Lot of raw materials consumed
    - e.g. Steel industry consumes large volumes of iron ore, coal, limestone, etc.
- ISO 9000 standards have many clauses corresponding to raw material control .
    - Not relevant to software organizations.

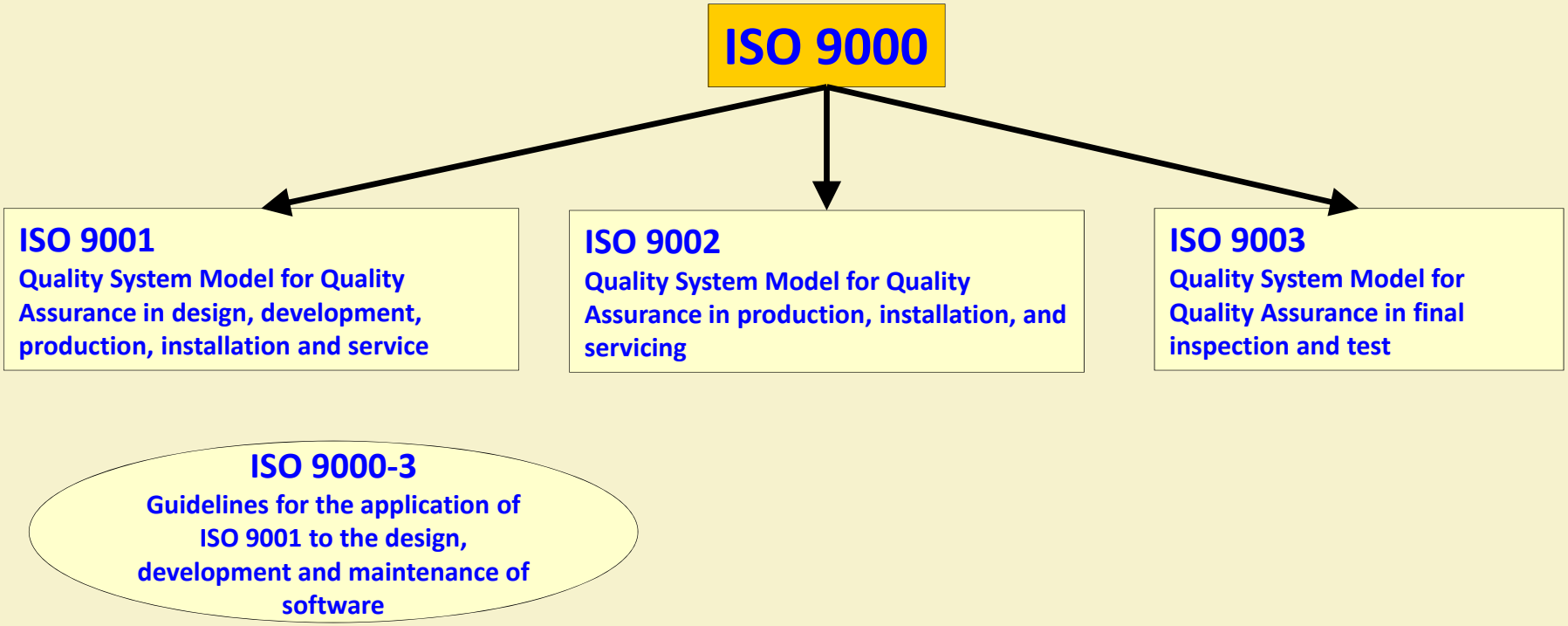
Software vs. Other Industries

- Radical differences exist between software and other product development:
  - Difficult to interpret various clauses of the original ISO standard in the context of software industry.

CONCEPTS COVERED

- ISO 9000

ISO 9000 Structure



ISO 9000 Part-3

- ISO released a separate document called ISO 9000 part-3 in 1991:
  - To help interpret the ISO standard for software industry.
- At present:
  - Official guidance is inadequate.

ISO 9000: 2000

- An important goal is to improve effectiveness via **process performance metrics:**
  - Numerical measurement of the effectiveness of various activities.
  - **Continual process improvement** and tracking customer satisfaction have been made explicit.

Why Get ISO 9000 Certification?

- Many software development contracts insist:
  - The Development organization to have ISO 9000 certification.

Why Get ISO 9000 Certification? Cont…

- Requires a well-documented software production process to be in place.
- **Contributes to repeatable and higher quality software.**
- Makes development process:
  - Focussed, efficient, and cost-effective

ISO does not certify…

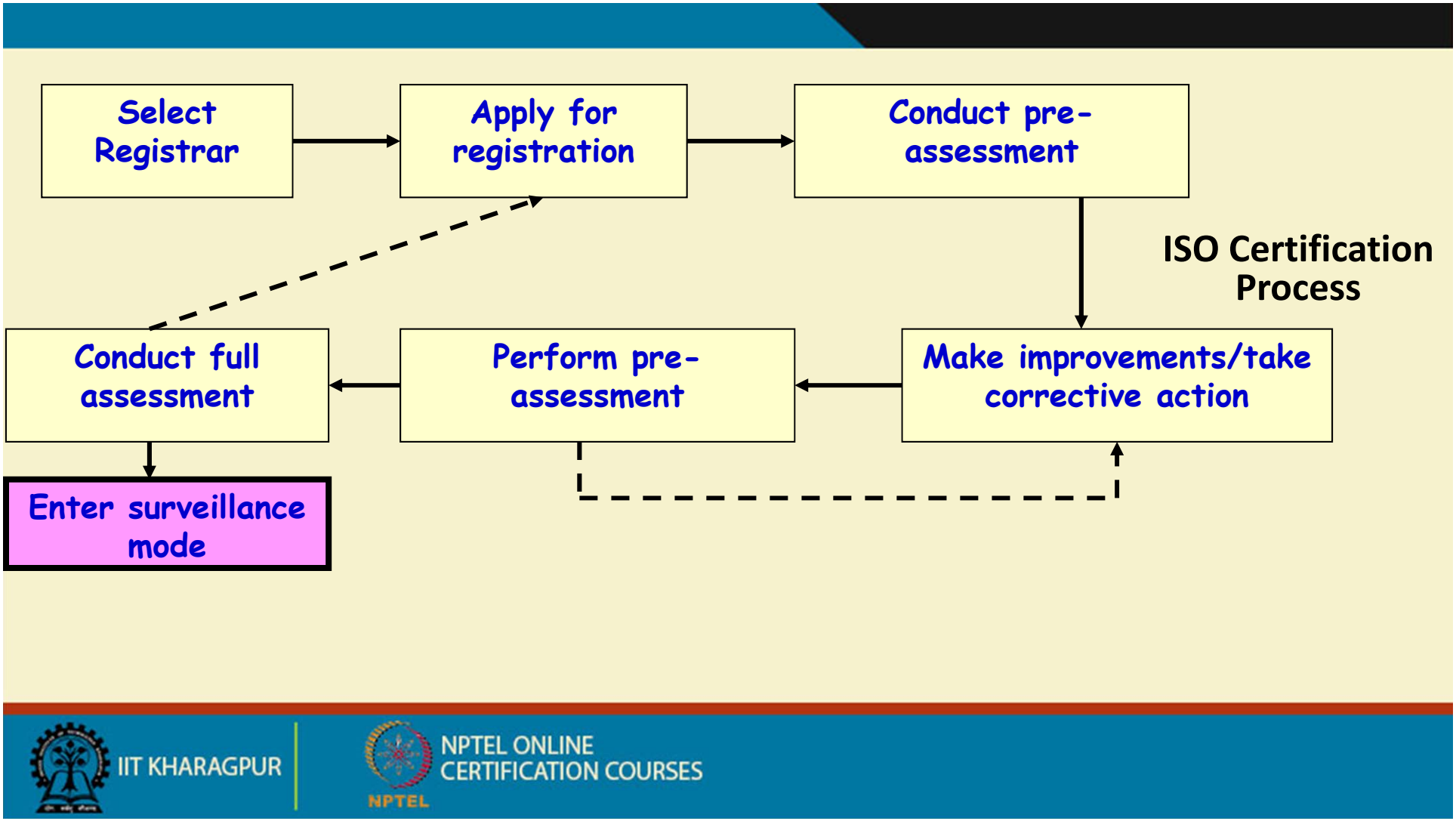
- ISO does not carry out ISO 9001 certification
- ISO does not issue certificates
- ISO does not accredit, approve or control the certification bodies
- ISO develops standards and guides to encourage good practice in accreditation and certification

Why Get ISO 9000 Certification? Cont…

- Points out the weakness of an organizations:
  - Recommends remedial action.
- Sets the basic framework:
  - For development of an optimal process and TQM.

How to Get ISO 9000 Certification?

- An organization intending to obtain ISO 9000 certification:
  - **Applies to an ISO 9000 registrar for registration.**
- ISO 9000 registration process consists of several stages.


Select Registrar
Apply for registration
Conduct pre-assessment
ISO Certification Process
Make improvements/take corrective action
Perform pre-assessment
Conduct full assessment
Enter surveillance mode

How to Get ISO 9000 Certification?

- **Application stage:**
  - Applies to a registrar for registration.
- **Pre-assessment:**
  - The registrar makes a rough assessment of the organization.

How to Get ISO 9000 Certification?

- **Document review and adequacy audit:**
  - Process and quality-related documents.
  - The registrar reviews the documents.
  - Makes suggestions for improvements.

How to Get ISO 9000 Certification?

- **Compliance Audit:**
- The registrar checks:
  - **Whether the suggestions made by it during review have been complied.**

How to Get ISO 9000 Certification?

- **Registration:**
  - The registrar awards ISO 9000 certificate after successful completion of all previous phases.
- **Continued surveillance:**
  - The registrar continues monitoring the organization periodically.

ISO 9000 Certification

- An ISO certified organization :
  - Can use the certificate for corporate advertizements.
  - Cannot use the certificate to advertize its products.
    - ISO 9000 certifies organization's process
    - Not any product of the organization.
  - An organization using ISO certificate for product advertizements:
    - Risks withdrawal of the certificate.

ISO 9001 Requirements

- Management responsibility
- Quality system
- Contract review
- Design Control
- Document and data control
- Purchasing
- Control of customer supplied product
- Product identification and traceability
- Process control
- Inspection and testing
- Control of inspection, measuring and test equipment

ISO 9001 Requirements cont…

- Inspection and test
- Control of non-confirming product
- Corrective and preventive action
- Handling, storage, packaging, preservation and delivery
- Control of quality records
- Internal quality audits
- Training
- Servicing
- Statistical techniques

Salient Features of ISO 9001 Requirements

- All documents concerned with the development of a software product:
  - **Should be properly managed, authorized, and controlled.**
- Proper plans should be prepared:
  - Progress against these plans should be monitored.

Salient Features of ISO 9001 Requirements

- Important documents independently checked and reviewed:
  - For effectiveness and correctness.
- The product should be tested :
  - Against specification.
- Several organizational aspects:
  - e.g., management reporting of the quality team.

Shortcomings of ISO 9001 Certification

- ISO 9000 requires a production process to be adhered to, however:
  - **Does not guarantee the process to be of high quality.**
  - **Does not give any guideline for defining an appropriate process.**

Shortcomings of ISO 9001 Certification cont...

- ISO 9000 certification process:
  - Not fool-proof
  - No international accredition agency exists.
  - Likely variations in the norms of awarding certificates:
    - Among different accredition agencies and among the registrars.

SEI Capability Maturity Model (CMM)

- Developed by Software Engineering Institute (SEI) of the Carnegie Mellon University, USA:
  - To assist the U.S. Department of Defense (DoD) in software acquisition.
- The rationale was:
  - **Include likely contractor performance as a factor in contract awards.**

SEI Capability Maturity Model

- Can be used in two ways:
  - **Capability evaluation**
  - **Software process assessment.**

Capability Evaluation

- Provides a way to assess the software process capability of an organization:
  - **Helps in selecting a contractor**
  - **Indicates the likely contractor performance.**

Software Process Assessment

- Used by an organization to assess its current process:
  - Suggests ways to improve the process capability.
  - **This type of assessment is for purely internal use.**

SEI Capability Maturity Model

- Major DoD contractors began CMM-based process improvement initiatives:
  - As they vied for DoD contracts.
- SEI CMM helped organizations:
  - Helped Improve quality of software they developed
  - **Realized adoption of SEI CMM model had significant business benefits.**
- Other organizations adopted CMM.

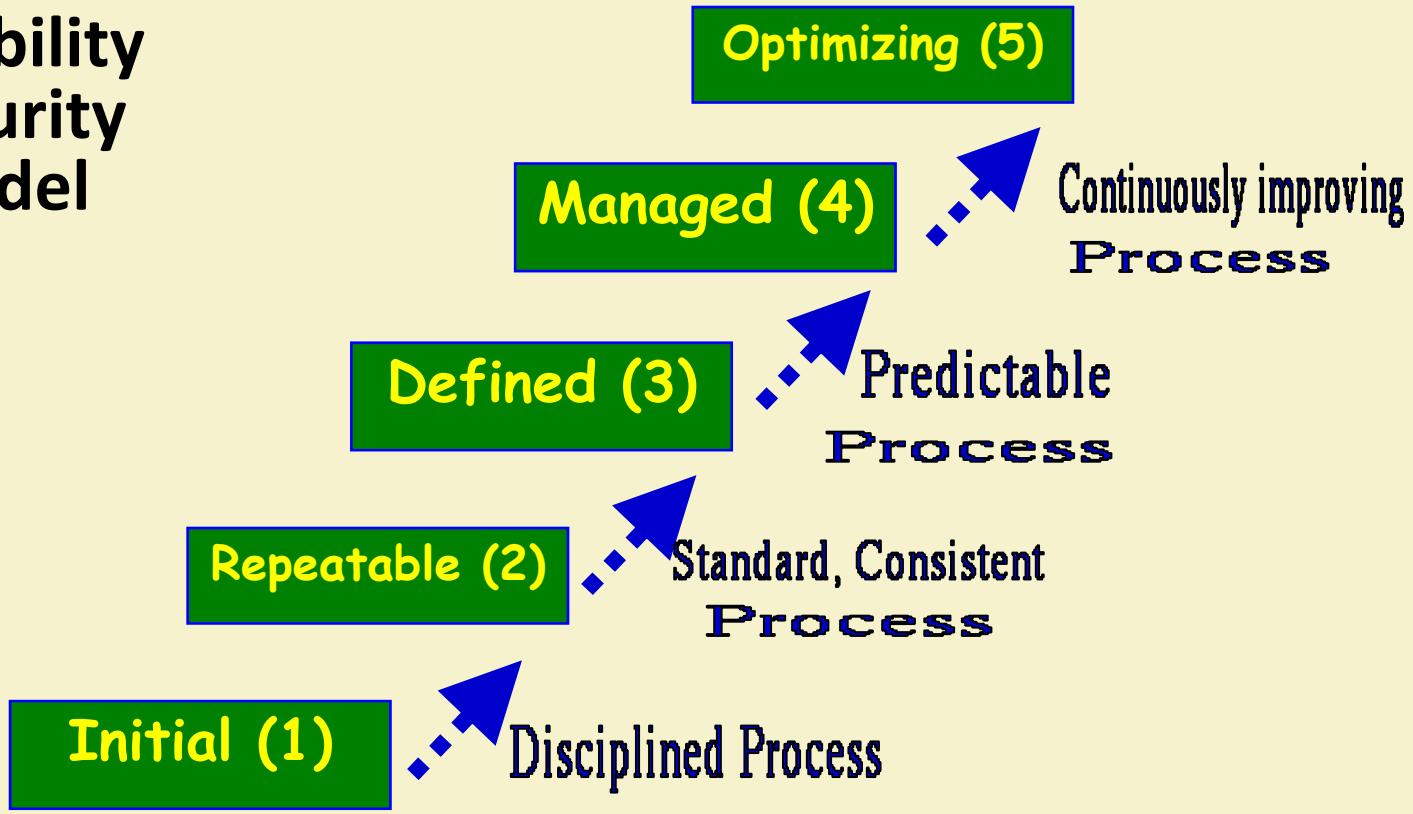
SEI Capability Maturity Model

- In simple words:
  - **CMM is a model for apprising the software process maturity of a contractor into different levels.**
  - Can be used to predict the most likely outcome to be expected:
    - from the next project that the organization undertakes.

Capability Maturity Model

- The SEI CMM classifies software development organizations into:
  - Five maturity levels.
  - Stages are ordered so that improvements at one stage provide foundations for the next.
  - Based on the pioneering work of Philip Crosby.

Capability Maturity Model



Level 1: (Initial)

- Organization operates
  - Without any formal process or project plans
- An organization at this level is characterized by
  - **Ad hoc and chaotic activities.**

Level 1 – Initial

- Most SW development organizations are at Level 1.
- Success comes from smart and hardworking people doing the right things
- Hard to recover from good people leaving
- Frequent crises and "firefighting."

Level 1: (Initial)

- Software development processes are not defined,
  - **Different engineers follow their own process**
  - The success of projects depend on individual efforts and heroics.

Level 1

- Schedules and budgets frequently missed
- Progress not measured
- Product configuration not controlled
- Engineering activities nonstandard, inconsistent
- Defects proliferate

Level 2: (Repeatable)

- Basic project management practices
  - Tracking cost and schedule exist.
- **Size and cost estimation techniques:**
  - **Function point analysis, COCOMO, etc. used.**
- Development process is ad hoc:
  - Not formally defined
  - Also not documented.

Level 2: (Repeatable)

- Process used for different projects might vary between projects:
  - **Earlier success on projects with similar applications can be repeated.**
  - Opportunity to repeat process exist when a company produces a family of products.

Level 3: (Defined)

- All management and development activities:
  - **Defined and documented.**
  - Common organization-wide understanding of activities, roles, and responsibilities.

Level 3: (Defined)

- The process though defined:
  - Process and product qualities are not measured.
- ISO 9001 aims at achieving this level.

Level 4: (Managed)

- Quantitative quality goals for products are set.
- Software process and product quality are measured:
  - The measured values are used to control the product quality.
  - Results of measurement used to evaluate project performance:
    - Rather than improve process.

Level 4: Managed

- Measurements of the software process and product quality are collected.
- Both the software process and products are quantitatively controlled.
  - **Variations in process performance narrowed to fall within acceptable quantitative bounds**
  - **Quantifiable and predictable**
    - **Quantitative quality goals are set**

Level 5: (Optimizing)

- Statistics collected from process and product measurements are analyzed:
  - **Continuous process improvement based on the measurements.**
    - Known types of defects are prevented from recurring by tuning the process
    - Lessons learned from specific projects incorporated into the process

Level 5: (Optimizing)

- Identify best software engineering practices and innovations:
  - Tools, methods, or process are identified.
  - Transferred throughout the organization.

CONCEPTS COVERED

- SEI CMM: KPAs
- SEI CMM vs. ISO 9001
- PSP

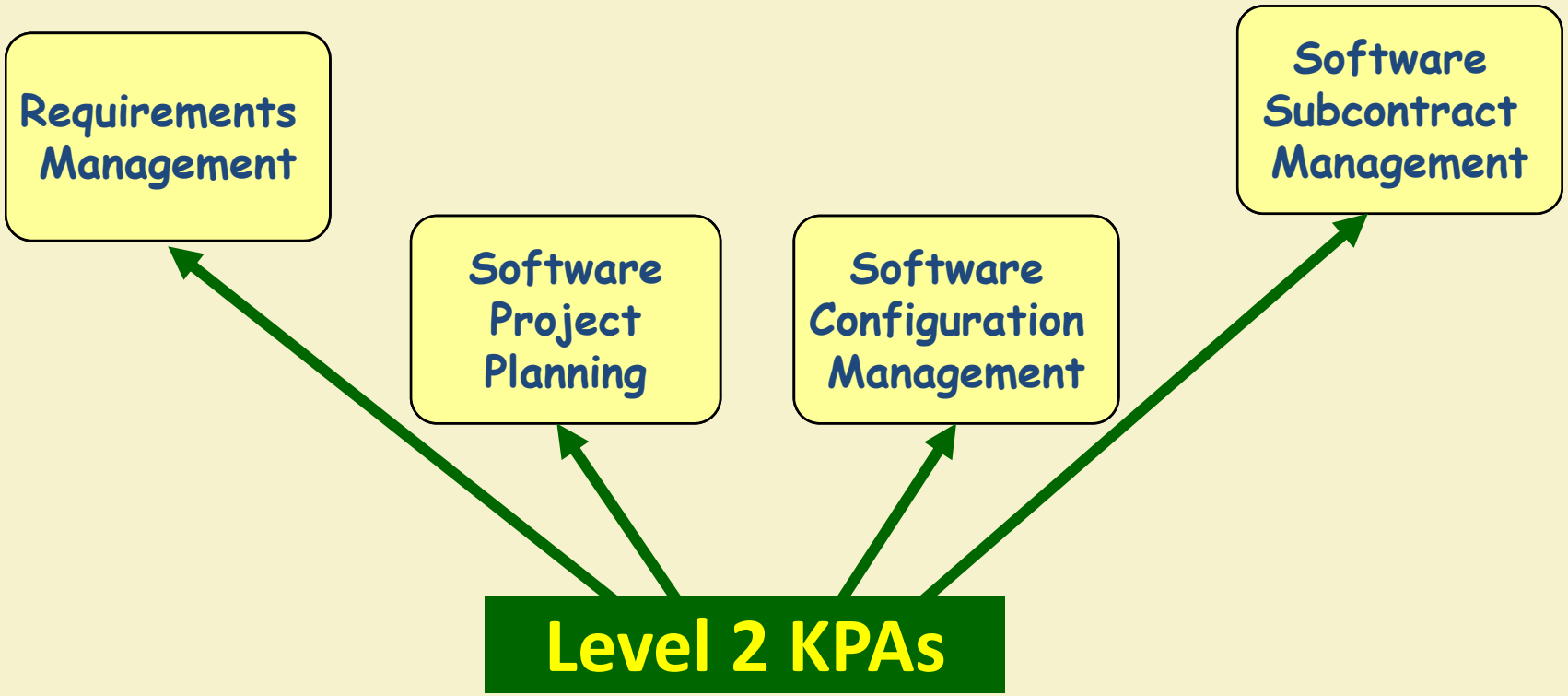


Key Process Areas

- Each level is associated with a key process area (KPA). It identifies:
  - **Process areas that an organization at the lower level must focus to reach this level.**

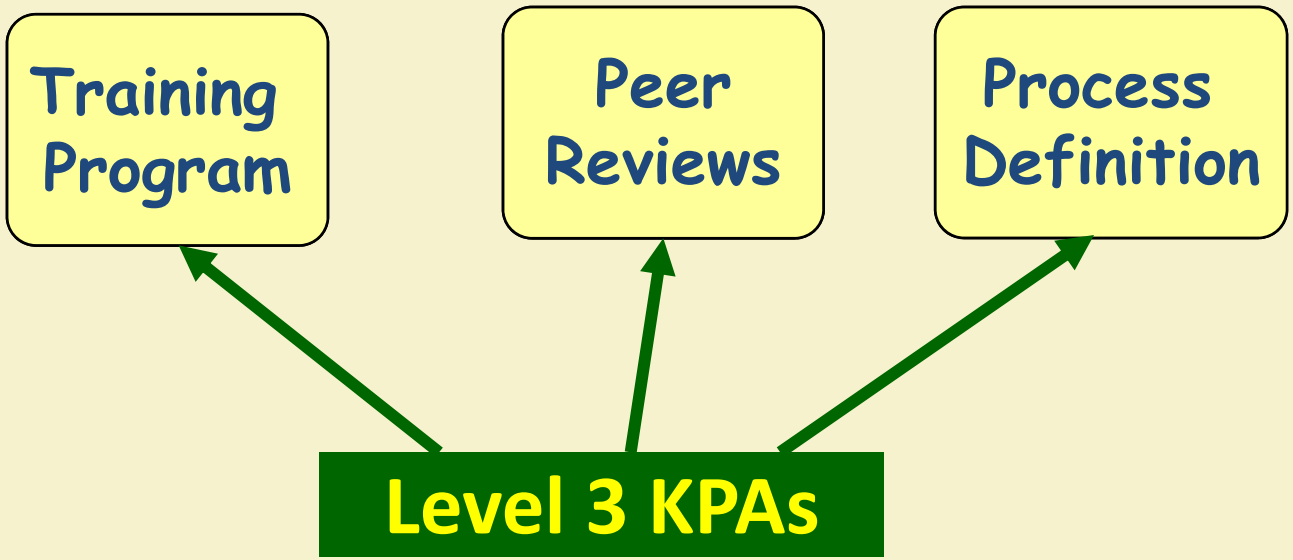
Level 2 KPAs

- **Requirements Management**
  - Requirements is the basis for planning and managing a software project
- **Software project planning:**
  - Size, cost, schedule.
  - Project monitoring
- **Configuration management**
- **Subcontract management**


Requirements Management
Software Project Planning
Software Configuration Management
Software Subcontract Management
Level 2 KPAs

Level 3 KPAs

- Process definition and documentation.
- Reviews
- Training program


Training Program
Peer Reviews
Process Definition
Level 3 KPAs

Level 4 KPAs

- **Quantitative measurements.**
- **Quantitative Process management.**
  - Control process performance quantitatively
  - Focus on identifying and correcting causes of variation

Level 5 KPAs

- **Defect prevention.**
  - Causal analysis of defects to prevent recurrence
- **Technology change management.**
  - Identify and transfer beneficial new technologies
    - tools
    - methods
    - processes
- **Process change management.**

Examples of metrics deployed

- Estimated source lines of code (SLOC)
- Actual SLOC
- Number of issues raised during code inspection
- Number of defects detected during unit testing
- Number of defects detected during system testing

Comparison Between ISO 9001 and SEI CMM

- ISO 9001 awarded by an international standards body:
  - Can be quoted in official documents and communications.
- SEI CMM assessment is purely for internal or limited use.

Comparison Between ISO 9001 and SEI CMM

- SEI CMM was developed specifically for software industry:
  - Addresses many issues specific to software industry.
- SEI goes beyond quality assurance
  - Aims for TQM.
  - ISO 9001 corresponds to SEI level 3.

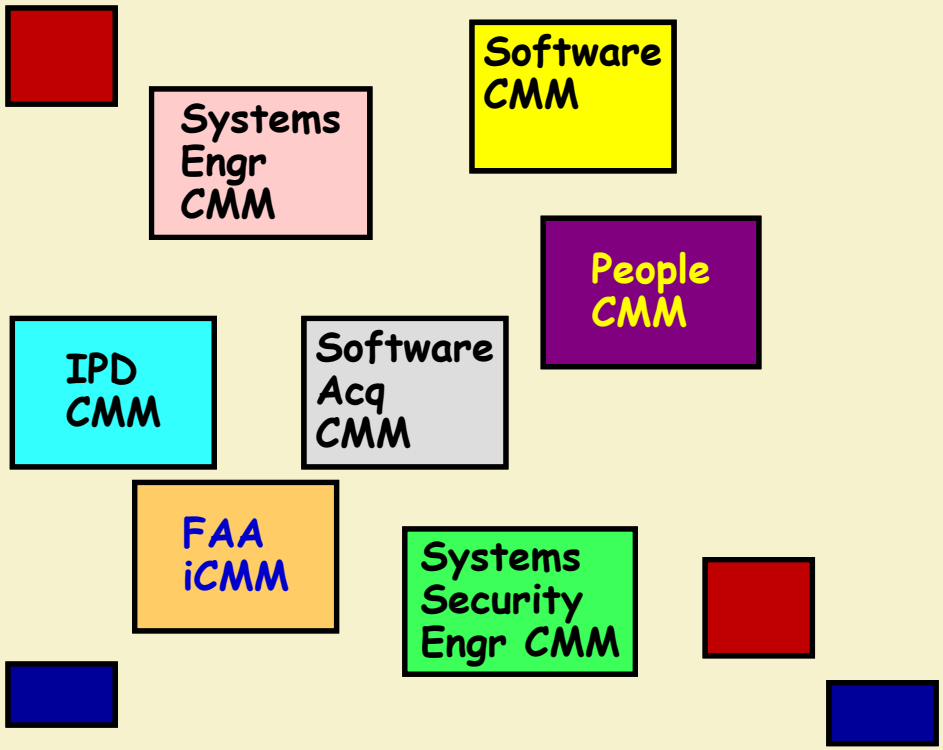
Comparison Between ISO 9001 and SEI CMM

- SEI CMM provides KPAs:
  - Process areas on which to focus to take an organization from one level to the other
- **Provides a way for gradual quality improvements over five stages.**
  - e.g trying to implement a defined process before a repeatable process:
    - Counterproductive as managers are overwhelmed by schedule and budget pressure.

Comparison of ISO and CMM

ISO	CMM
Is a Certification	Is an assessment
Used for all industry development	Used for software
Yearly re-certification	No follow up after reaching level
Single level	Five levels
Third Party Certification	No certification

Proliferation of CMMs…



- Different structures, terms, ways of measuring maturity
- Causes confusion
- Hard to use for a combined improvement program
- Hard to use multiple models for a supplier selection


Thank you